

## BUDAORS, Hungary

WMO: 128380

Lat: 47.450N

Lon: 18.967E

Elev: 433

StdP: 14.47

Time Zone: 1.00 (EUC)

Period: 82-92

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 11.8       | 15.8 | 6.1                           | 7.7 | 14.1 | 9.9 | 9.3 | 18.5 | 40.2                  | 30.4 | 37.1 | 33.3 | 4.2                   | 270  | 0.496 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.5                   | 87.8            | 68.2 | 84.8 | 67.4 | 82.0 | 66.5 | 70.3                | 84.0 | 68.9 | 81.5 | 67.6 | 79.0 | 6.5                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 65.5                            | 95.8 | 75.8 | 64.3 | 91.6 | 74.6 | 63.1 | 87.7 | 73.3 | 34.4          | 84.0 | 33.2 | 82.2 | 32.2 | 78.9 | 75.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 31.2          | 26.0 | 20.8 | 2.7                        | 92.4 | 9.1                | 2.5 | -3.9                                               | 94.1 | -9.2       | 95.6 | -14.4      | 97.0 | -21.0      | 98.8 |    |    |
| (5)               |      |      | 3.1                        | 73.5 | 7.6                | 1.3 | -2.4                                               | 74.4 | -6.9       | 75.2 | -11.2      | 75.9 | -16.8      | 76.9 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 51.1   | 31.5  | 33.7  | 42.0  | 52.1  | 61.1  | 64.8  | 70.0  | 68.9  | 62.3  | 51.3  | 39.2  | 34.6  |  |  |
|                                                                        | DBStd       | 15.35  | 8.74  | 8.39  | 8.04  | 6.24  | 5.89  | 5.47  | 5.38  | 5.75  | 5.51  | 7.16  | 6.83  | 7.52  |  |  |
|                                                                        | HDD50       | 2224   | 574   | 456   | 265   | 49    | 4     | 0     | 0     | 0     | 1     | 69    | 328   | 478   |  |  |
|                                                                        | HDD65       | 5530   | 1039  | 875   | 713   | 388   | 146   | 68    | 17    | 29    | 115   | 424   | 775   | 941   |  |  |
|                                                                        | CDD50       | 2614   | 0     | 1     | 17    | 112   | 348   | 444   | 619   | 586   | 371   | 111   | 3     | 2     |  |  |
|                                                                        | CDD65       | 443    | 0     | 0     | 0     | 1     | 24    | 62    | 171   | 150   | 35    | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 3760   | 0     | 0     | 0     | 7     | 181   | 497   | 1548  | 1256  | 268   | 3     | 0     | 0     |  |  |
|                                                                        | CDH80       | 1032   | 0     | 0     | 0     | 0     | 25    | 99    | 489   | 380   | 39    | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 6.7    | 7.2   | 7.1   | 7.4   | 8.1   | 6.4   | 7.1   | 6.8   | 5.9   | 6.0   | 5.5   | 6.2   | 7.4   |  |  |
|                                                                        | PrecAvg     | 22.0   | 1.5   | 1.3   | 1.3   | 1.7   | 2.3   | 2.6   | 1.9   | 2.0   | 1.6   | 1.5   | 2.4   | 1.8   |  |  |
| (15) Precipitation                                                     | PrecMax     | 33.1   | 3.4   | 5.4   | 2.3   | 3.8   | 5.8   | 9.4   | 5.0   | 5.4   | 4.6   | 6.1   | 6.8   | 4.7   |  |  |
| (16) PrecMin                                                           |             | 12.9   | 0.0   | 0.0   | 0.1   | 0.4   | 0.0   | 0.3   | 0.1   | 0.0   | 0.0   | 0.0   | 0.4   | 0.0   |  |  |
| (17) PrecStd                                                           |             | 4.8    | 0.9   | 1.1   | 0.7   | 0.9   | 1.4   | 1.8   | 1.2   | 1.3   | 1.2   | 1.4   | 1.6   | 1.1   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 50.9  | 61.4  | 70.4  | 75.7  | 83.8  | 86.5  | 91.8  | 90.3  | 84.7  | 73.5  | 59.9  | 57.9  |  |  |
|                                                                        |             | MCWB   | 44.6  | 50.3  | 55.4  | 59.6  | 63.8  | 68.3  | 68.6  | 68.8  | 66.5  | 61.0  | 52.9  | 51.4  |  |  |
|                                                                        | 2%          | DB     | 47.6  | 53.6  | 62.5  | 71.1  | 78.6  | 82.8  | 88.5  | 87.5  | 80.5  | 69.5  | 53.6  | 52.2  |  |  |
|                                                                        |             | MCWB   | 43.4  | 46.2  | 51.0  | 56.6  | 62.8  | 68.2  | 68.9  | 67.5  | 64.6  | 59.3  | 48.6  | 47.9  |  |  |
|                                                                        | 5%          | DB     | 45.3  | 47.9  | 58.3  | 67.3  | 75.4  | 79.6  | 85.5  | 84.4  | 77.2  | 65.8  | 50.5  | 48.4  |  |  |
|                                                                        |             | MCWB   | 41.7  | 42.3  | 49.2  | 55.4  | 62.1  | 66.0  | 68.0  | 67.1  | 64.1  | 56.5  | 46.5  | 44.4  |  |  |
|                                                                        | 10%         | DB     | 42.3  | 43.9  | 54.3  | 63.5  | 72.4  | 76.4  | 82.3  | 81.3  | 73.6  | 62.4  | 48.5  | 44.9  |  |  |
|                                                                        |             | MCWB   | 39.2  | 40.0  | 46.7  | 53.5  | 60.6  | 64.2  | 66.6  | 66.0  | 62.1  | 54.7  | 45.4  | 41.7  |  |  |
|                                                                        | 0.4%        | WB     | 45.6  | 51.1  | 56.9  | 60.7  | 66.2  | 71.0  | 72.8  | 71.5  | 67.6  | 62.6  | 53.9  | 52.2  |  |  |
|                                                                        |             | MCDB   | 49.7  | 60.4  | 68.5  | 72.7  | 77.8  | 82.9  | 85.8  | 84.7  | 79.4  | 70.2  | 57.5  | 58.7  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 43.8  | 46.6  | 52.2  | 57.8  | 64.4  | 69.2  | 70.5  | 69.9  | 66.3  | 60.0  | 49.2  | 47.8  |  |  |
|                                                                        |             | MCDB   | 47.1  | 52.8  | 60.6  | 68.4  | 75.7  | 80.5  | 84.9  | 84.0  | 78.4  | 67.9  | 51.9  | 51.6  |  |  |
|                                                                        | 5%          | WB     | 42.1  | 42.9  | 49.8  | 56.0  | 62.9  | 67.3  | 69.1  | 68.4  | 64.7  | 57.9  | 47.4  | 45.0  |  |  |
|                                                                        |             | MCDB   | 45.0  | 46.8  | 57.0  | 65.6  | 72.7  | 77.2  | 82.2  | 81.2  | 74.9  | 64.2  | 49.7  | 48.4  |  |  |
|                                                                        | 10%         | WB     | 39.5  | 40.4  | 47.4  | 54.1  | 61.6  | 65.4  | 67.8  | 67.0  | 62.9  | 55.8  | 45.7  | 41.9  |  |  |
|                                                                        |             | MCDB   | 42.1  | 43.7  | 53.6  | 62.3  | 70.5  | 74.4  | 79.4  | 78.4  | 71.7  | 60.9  | 48.2  | 44.7  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 9.0   | 11.3  | 14.5  | 15.8  | 16.4  | 16.1  | 18.5  | 18.2  | 18.3  | 16.5  | 11.5  | 8.7   |  |  |
|                                                                        |             | MCDBR  | 12.1  | 19.5  | 22.3  | 22.3  | 22.2  | 22.2  | 24.1  | 24.1  | 23.3  | 22.1  | 14.8  | 12.3  |  |  |
|                                                                        |             | MCWBR  | 8.6   | 12.8  | 13.2  | 11.2  | 9.4   | 9.5   | 8.5   | 8.9   | 10.0  | 12.4  | 10.4  | 8.6   |  |  |
|                                                                        | 5% WB       | MCDBR  | 10.8  | 16.8  | 20.4  | 19.8  | 18.9  | 20.3  | 21.4  | 21.0  | 22.0  | 19.4  | 12.4  | 12.6  |  |  |
|                                                                        |             | MCWBR  | 8.1   | 11.6  | 12.7  | 10.7  | 8.6   | 9.6   | 8.2   | 8.4   | 10.4  | 11.3  | 9.7   | 9.1   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.331 | 0.354 | 0.391 | 0.439 | 0.429 | 0.429 | 0.442 | 0.442 | 0.417 | 0.385 | 0.360 | 0.330 |  |  |
|                                                                        | taud        |        | 2.363 | 2.315 | 2.241 | 2.123 | 2.210 | 2.243 | 2.209 | 2.221 | 2.256 | 2.324 | 2.349 | 2.373 |  |  |
|                                                                        | Ebn at Noon |        | 234   | 253   | 260   | 258   | 266   | 266   | 260   | 254   | 249   | 238   | 219   | 219   |  |  |
|                                                                        | Edh at Noon |        | 23    | 30    | 37    | 46    | 43    | 43    | 43    | 41    | 36    | 29    | 23    | 21    |  |  |
|                                                                        | RadAvg      |        | 329   | 587   | 983   | 1450  | 1750  | 1899  | 1860  | 1668  | 1173  | 751   | 396   | 249   |  |  |
| (45) All-Sky Solar Radiation                                           | RadStd      |        | 39    | 86    | 146   | 139   | 183   | 144   | 133   | 161   | 138   | 81    | 38    | 33    |  |  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (51 neighbors) | +1.08 | N/A     | N/A    | N/A     | +0.94 | +0.90 | -221        | -322  | +182  | +97   |

Nomenclature: See separate page